

RV College of Engineering®, Bengaluru – 59
Department of Information Science and Engineering
DATABASE MANAGEMENT SYSTEMS (CD252IA)

Synopsis

TITLE: Farming Resource Sharing System		
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1. Introduction

The Farming Resource Sharing System is a collaborative platform designed to help farmers with diverse expertise—such as machinery operations, organic farming, and mixed cropping—work together effectively. The system streamlines collaboration by aligning farmers’ skills, interests, and resources to maximize land productivity and achieve higher agricultural outcomes. It also provides a centralized platform for communication, resource management, and coordinated decision-making, enabling more efficient and sustainable farming practices.

The application is developed using **React (with Vite)** for a fast and responsive frontend, while a **Node.js backend** handles server-side logic and API management. The system uses **Neon Cloud PostgreSQL** as the database for storing farmer profiles, land details, machine availability, collaboration requests, and participant information. By enforcing constraints such as **resource availability limits, role-based participation (e.g., machinery, organic farming, mixed cropping), and controlled collaboration groups**, the platform promotes fair resource sharing, effective coordination, and improved agricultural outcomes through skill-based collaboration among farmers.

2. Existing System

The existing system in farming is largely **manual and fragmented**, where farmers rely on personal contacts, local agents, or informal groups to share resources and information. Equipment lending, land use planning, and collaboration are usually managed through phone calls or word of mouth, with no centralized digital platform. This results in inefficient resource utilization, limited collaboration, and reduced productivity.

3. Proposed System

The Farming Resource Sharing System proposes an automated, matchmaking platform that enables farmers to collaborate effectively based on their expertise and resources.

Key features include:

1. User Authentication and Profiles

- Users can register, log in, and create profiles with details such as name, phone number, and email address.

2. Group Joining and Description

- Farmers can view all available farming groups on the platform.
- Each group shows a brief summary of the resources involved, helping farmers choose groups based on their interests.

3. Farmers and Land Display

- Details and contact information of all registered farmers are displayed to support collaboration.
- Farmers can add land information, including soil type, land area, and location.

4. Equipment Lending and Management

- Farmers can list the machinery and tools they own and mark them as available for sharing.
- They can view lending requests and directly contact farmers who require the equipment.

5. Crop Logging Feature

- Farmers can log the crops they are cultivating along with expected yield and current growth stage

4. Relational Database Structure

1. Farmers Table

- Stores information about all registered farmers using the platform.
- Each farmer is uniquely identified using a Farmer ID (Primary Key).
- Includes authentication reference through Clerk User ID, along with name, phone number, and address.
- Used for farmer identification, communication, and authorization within the system.

- A farmer can participate in multiple farming groups.
- A farmer can own land, equipment, and grow crops.

Cardinality:

- One farmer can belong to many farmer groups.
 - One farmer can own many land records, equipment items, and crops.
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2. Farmer_Groups Table

- Stores details of all farming collaboration groups available on the platform.
- Each group is uniquely identified using a **Group ID (Primary Key)**.
- Contains group name and description explaining the collaboration purpose.
- Used to organize farmers based on shared farming goals, skills, and resources.
- A group can include multiple farmers.
- A group can be associated with land and shared equipment.

Cardinality:

- One farmer group can have many farmers, land records, and equipment items.
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3. Farmer_Group_Members Table

- Represents the association between farmers and farming groups.
- Resolves the **many-to-many relationship** between Farmers and Farmer_Groups.
- Stores **Farmer ID (Foreign Key)** and **Group ID (Foreign Key)**.
- Tracks the date when a farmer joined a particular group.
- Ensures proper membership management and group participation tracking.

Cardinality:

- Many farmers can join many farmer groups through this table.
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4. Land Table

- Stores details of agricultural land owned by farmers.
- Each land record is uniquely identified using a **Land ID (Primary Key)**.
- Linked to a farmer using **Farmer ID (Foreign Key)**.
- May also be associated with a farming group using **Group ID (Foreign Key)**.
- Contains land area, location, and soil type information.
- Used to manage land resources and support crop planning.

Cardinality:

- One farmer can own many land records.

- One farmer group can be associated with many land records.
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5. Equipment Table

- Stores information about farming equipment and machinery.
- Each equipment item is uniquely identified using an **Equipment ID (Primary Key)**.
- Linked to the owner farmer using **Farmer ID (Foreign Key)**.
- Can be associated with a farming group using **Group ID (Foreign Key)**.
- Includes equipment name, type, and availability status.
- Used to support equipment sharing and lending among farmers.

Cardinality:

- One farmer can own many equipment items.
 - One farmer group can share many equipment items.
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6. Crops Table

- Stores information about crops grown by farmers.
- Each crop record is uniquely identified using a **Crop ID (Primary Key)**.
- Linked to land using **Land ID (Foreign Key)**.
- Linked to the farmer using **Farmer ID (Foreign Key)**.
- Contains crop name, growth stage, and expected yield details.
- Used to track crop progress and assist in production planning.

Cardinality:

- One land record can have many crops.
 - One farmer can grow many crops.
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5. Societal Concern

The project addresses important **social concerns** by promoting collaboration, resource sharing, and inclusivity among farmers. By enabling farmers to share land, equipment, and expertise through a centralized digital platform, the system helps reduce inequality between small and large farmers and minimizes resource wastage. It also encourages sustainable farming practices, improves rural connectivity, and supports better livelihoods by empowering farmers with access to shared knowledge and resources.

How EL Team Finder Helps

- Encourages collaboration among farmers with different skills and expertise.
- Enables efficient sharing of farming resources such as land and equipment.
- Reduces operational costs by minimizing the need to purchase expensive machinery.
- Improves land utilization and increases overall agricultural productivity.
- Supports small and marginal farmers by providing access to shared resources.
- Promotes sustainable and organized farming practices.
- Provides a centralized platform for communication and coordination.